

LAHORE ALLAMA IQBAL INTL, Pakistan

WMO: 416410

Lat: 31.522N

Lon: 74.404E

Elev: 712

StdP: 14.32

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	37.8	40.8	26.3	20.8	64.5	30.1	24.8	61.9	14.0	59.8	12.2	61.3	1.2	270	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	21.2	109.7	74.0	107.4	73.9	104.2	73.9	84.4	94.2	83.5	92.9	82.6	91.6	8.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.4	173.7	90.0	81.0	165.1	88.9	80.5	162.4	88.6	49.4	93.8	48.2	93.0	47.2	92.0	95.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
17.9	14.3	12.1	34.2	114.8	3.2	2.1	31.9	116.3	30.0	117.5	28.2	118.7	25.9	120.2		
			33.0	86.5	3.2	3.0	30.7	88.6	28.8	90.4	27.0	92.1	24.7	94.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	75.9	53.6	60.7	70.4	81.4	90.2	91.5	87.9	86.9	84.8	78.1	66.8	57.1
	DBStd	13.86	4.30	4.65	5.91	6.20	5.63	5.70	4.11	3.51	3.52	4.83	4.19	4.47
	HDD50	21	16	1	0	0	0	0	0	0	0	0	0	4
	HDD65	780	354	134	18	0	0	0	0	0	0	0	28	246
	CDD50	9455	127	299	633	942	1247	1245	1176	1144	1043	871	504	224
	CDD65	4741	0	13	186	493	782	795	711	679	593	406	82	1
	CDH74	65638	6	168	1968	6436	12297	12668	10172	9176	7399	4380	913	55
	CDH80	38363	0	17	765	3771	8301	8568	5888	4948	3806	2117	182	0
Wind	WSAvg	4.7	3.4	4.7	5.2	5.8	6.4	6.9	6.3	5.1	4.0	3.1	2.6	2.8
Precipitation	PrecAvg	21.7	0.7	0.7	1.5	0.6	0.7	1.2	7.8	6.1	2.5	0.3	0.1	0.5
	PrecMax	35.4	3.9	3.1	6.1	2.7	2.3	3.3	26.2	19.6	6.5	1.3	0.6	3.2
	PrecMin	10.2	0.0	0.0	0.0	0.0	0.0	0.0	1.7	1.1	0.0	0.0	0.0	0.0
	PrecStd	7.9	0.9	0.8	1.6	0.7	0.7	1.1	5.4	5.0	2.0	0.4	0.2	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.8	82.4	95.3	107.6	113.2	113.8	106.0	100.7	99.0	96.9	87.5	78.6
		MCWB	55.4	60.8	67.9	70.2	74.0	73.9	80.2	80.6	77.5	74.2	64.0	58.4
	2%	DB	71.2	78.7	91.4	103.9	109.8	110.9	102.0	98.6	97.1	94.9	84.3	75.4
		MCWB	54.7	59.3	67.0	69.3	73.2	74.5	80.8	80.9	77.8	72.9	64.0	57.5
	5%	DB	68.1	75.4	87.8	100.4	107.6	107.6	98.7	96.7	95.3	93.0	82.2	73.1
		MCWB	53.5	58.7	65.7	69.1	73.2	75.1	81.2	80.8	77.9	71.9	62.7	56.4
	10%	DB	65.8	73.0	84.2	96.7	104.4	104.3	96.6	94.8	93.5	90.6	80.2	70.0
		MCWB	52.7	57.5	64.3	68.5	72.2	76.0	80.8	80.9	77.8	71.1	62.2	55.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	65.0	72.4	77.3	82.0	84.7	86.4	85.4	83.6	78.9	69.3	62.4
		MCDB	68.5	75.5	88.2	97.5	104.5	99.5	95.7	94.2	92.5	91.5	80.3	72.0
	2%	WB	57.5	62.6	70.0	73.6	79.2	83.0	84.6	84.2	82.1	77.0	67.1	60.3
		MCDB	66.3	73.5	85.3	95.7	100.9	97.3	93.8	92.6	91.0	89.3	78.6	70.5
	5%	WB	55.6	60.9	68.0	71.9	77.0	81.5	83.6	83.2	80.9	75.3	65.2	58.4
		MCDB	64.0	71.3	82.5	94.1	99.0	95.8	92.8	91.2	90.1	86.6	77.2	68.8
	10%	WB	54.1	59.1	66.2	70.2	75.2	80.3	82.7	82.4	79.8	73.7	63.5	56.8
		MCDB	62.7	69.3	80.5	90.5	98.6	95.2	91.5	90.0	88.9	85.4	75.9	67.4
Mean Daily Temperature Range	5% DB	MDBR	22.2	22.3	23.5	25.9	25.7	21.2	14.9	14.2	17.5	23.7	26.5	25.1
		MCDBR	26.9	26.0	27.6	30.1	28.4	26.2	18.0	16.6	19.2	24.0	27.9	28.5
		MCWBR	13.1	10.9	9.3	8.9	8.8	7.5	6.3	5.8	6.1	7.1	11.0	13.0
	5% WB	MCDBR	22.3	21.8	24.3	26.3	25.5	19.0	14.9	14.2	16.2	20.6	24.5	23.9
		MCWBR	12.9	10.9	10.2	9.9	10.4	8.1	7.1	6.6	7.1	7.4	11.0	12.6
Clear-Sky Solar Irradiance	taub	0.488	0.495	0.512	0.573	0.654	0.748	0.789	0.697	0.606	0.659	0.622	0.538	
	taud	1.855	1.891	1.880	1.789	1.652	1.527	1.499	1.657	1.788	1.596	1.616	1.755	
	Ebn at Noon	220	237	245	236	218	198	189	204	217	188	179	195	
	Edh at Noon	53	56	61	69	80	91	93	78	66	75	68	56	
All-Sky Solar Radiation	RadAvg	908	1249	1667	1992	2201	2046	1739	1675	1649	1459	1122	935	
	RadStd	105	128	105	89	76	118	109	128	88	72	61	57	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page